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1) In Cyclic Redundancy Checking, CRC is _____.

- Divisor
- Quotient
- **Remainder** **page 20**
- Dividend

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2) The satellite or radio topology in which all computers are connected to each other via satellite or radio wave is a kind of

- **Broadcast network** **page 5**
- Point-to-Point network

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3) Which of the following two operations are performed by Ethernet CD?

I. It monitors outgoing signals.
II. Grabbled signal is interpreted as a collision.
III. Signals are sent back to the sender.

- I and III
- II and IV
- II and III
- **I and II** **page 28**

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4) **The process of forwarding a packet is called _____.**

- **Routing** **page 58**
- Processing
- Hierarchical Addressing
- Source Addressing

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5) **Which statement is correct about Network Interface Card (NIC)?**

I. NIC contains sufficient hardware to process data independent of system CPU
II. NIC looks like any other I/O device to system CPU
III. NIC also receives interrupt on incoming data.

- I and II
- II and III

- I and III
- **All of the given options** **page 40**

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6) **While transmitting data from source A to destination B, 4 bits are changed during the transmission then the error is a _____.**

- **Burst error** **page 22**
- Single error
- Double error
- Logic error

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7) **The switch that has no attached computers is called _____.**

- Packet Switch
- Exterior Switch
- **Interior Switch** **page 59**
- External Switch

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8) **_____ is the most popular wiring scheme because of lowest cost.**

- 5 Base T
- **10 Base T** **page 43**
- 100 base T
- 1000 base T

---

9) **In _____ technique, all members of the network can send data only on the specific time slot?**

- CDMA
- FDMA
- **CSMA** **page 43**
- TDMA

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10) **If a sender with a 100 Megabit NIC and receiver with 10 Megabit NIC wants to communicate with each other at which speed they can communicate?**

- **100 Megabit** **page 40**
- 1000 Megabit
- 10 Megabit

11) **Missing _____ in a frame format indicates receiving computer missed beginning of the message.**

- eot
- sot
- eoh
- **soh** **page 16**

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12) **The _____ of errors is easier than the _____ of errors.**

- correction; detection
- correction; creation
- detection; manipulation
- **detection; correction** **cs601 handouts**

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13) **The _____ of errors is difficult than the _____ of errors.**

- detection; correction
- correction; creation
- detection; manipulation
- **correction; detection** **google**

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14) **Source Independence phenomenon allows fast and efficient _____.**

- Source Addressing
- Routing
- **Packet Switching** **page 58**
- Store and forward

---

15) **In distributed route computation process each packet switch computes its routing table locally and sends messages to the _____.**

- Router
- Graph
- **Neighbors** **page 63**
- WAN

---

16) **The ___ layer of the OSI model takes the data from the physical layer and perform error checking.**

- **Data Link** **Google**

- Transport
- Application
- Presentation

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17) CBR stands for _____.

- Constant Byte Rate
- **Constant Bit Rate**
- Connection break Recovery

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18) A network that uses _____ usually consist of a single long cable to which computer attach.

- Star Topology
- Ring Topology
- **Bus Topology** **page 26**
- Bus and Ring Topology

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19) If simultaneous transmission occurs, the frames interfere with each other and this phenomenon is called _____.

- packet loss
- **collision** **page 28**
- delayed transmission
- coordination

---

20) In _____ all computers are attached to a central hub

- Ring topology
- **Star topology** **page 31**
- Hub topology
- Bus topology

---

21) For adding a new computer in a point to point network _____new connection will be required

- N
- **N-1** **page 23**
- N-2

22) The fastest Ethernet hardware operates at a rate of _____

- 1000 bps
- **100 Mbps** **page 27**
- 10 Mbps
- 1000 Mbps

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23) Which of the following the benefit to point to point communication?

- Quick to configure.
- **Security implementation** **page 23**
- Cost
- Less availability

---

24) ATM offers _____speed and _____delay

- High, High
- **High, low** **page 72**
- Low, low
- Low, high

---

25) Thick Ethernet is known as _____

- 10 Base 2
- **10 Base 5** **google**
- 10 Base T
- 10 Base 10

---

26) Both interior and exterior switches _____packet and they also need routing tables.

- Create
- Delay
- **Forward** **page 59**
- Start

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27) _____is the rate at which data can be sent through the network

- **Throughput** **page 79**
- Propagation delay

- Access delay
- Switching delay

---

28) In a ping command, small –sized information packets of _____are sent.

- TCP
- UDP
- **ICMP** **page 11**
- IGMP

---

29) Many WANs use _____addressing of efficiency

- Sequence
- Forward
- **Hierarchical** **page 56**
- Sequential

---

30) Which of the following is not a function of ping commends?

- Local computer network issue
- Congestion in the network
- **Relay nodes** **page 11**
- Remote host availability

---

31) Packet switched can be linked together to form a _____

- LAN
- **WAN** **page 56**
- Bridge
- None of the given option

---

32) Fixed network is a type of network which is classified with repect to the factor

- Size
- Connectivity
- Medium
- **Mobility** **page 6**

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33) Parity bit can detect _____error(s)

- 4
- 3
- **2** **page 19 (even or odd)**
- 1

---

34) When we will try to connect to the internet which of the following protocol will be used?

- Telnet
- **TCP/IP** **page 84**
- ICMP
- Ipx

---

35) How many computers are mandatory to form a network?

- 1
- **2** **google**
- 4
- 8

---

36) _____ is standard wireless LAN that uses radio signals at 2.4GHz.

- IEEE 803.16
- IEEE 802.3
- **IEEE 802.11** **page 29**

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37) LAN interface hardware obeys _____ access rule.

- **CSMA/CD** **page 34**
- Round robin
- First come first served
- High priority

---

38) The most important task a bridge performs is _____.

- Packet switching
- Line controlling
- **Frame filtering** **page 50**

---

39) ATM uses small, fixed-sized packets called cells and each cell has _____ octets.

- 47
- 52
- **53** **page 66**
- 55

---

40) _____ can't be guaranteed with variable length packets as easily as it can with fixed length cells.

- Errors
- Noise
- Quality of Service
- **None of the given** **page 72**

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41) Which multiplexing technique transmits digital signals?

- WDM
- FDM
- **TDM** **page 15**
- GSM

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42) Connections in Asynchronous transfer mode are called _____.

- **Virtual channels** **page 67**
- Virtual Path Identifier
- Cells
- Cell tax

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43) A _____ is a device that combines multiple signals and forwards into a single line.

- Modem
- **Multiplexer** **page 41**
- Repeater
- Layer 3 Switch

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44) _____ is inflexible.

- **Static routing** **page 62**

- Processing
- Searching
- Routing

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45) If a hub is used to connect many computers in a LAN, then only _____ computers can communicate at a given time.

- Three
- **Two** **page 54**
- Four
- One

---

46) _____ is the term used for variance in transmission delays.

- Friction
- Segment
- Checksum
- **Jitter** **page 66**

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47) _____ is also called self healing network.

- ATM
- FDDI
- Token Ring
- **Frame relay** **page 46**

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48) _____ and _____ are the two frame types.

- Internal and external
- Single and dual
- Implicit and explicit
- **Multicast and broadcast** **page 35**

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49) Bluetooth uses shared medium and _____ instead of coaxial cable.

- **Radio waves** **page 29**
- Infrared waves
- Micro waves
- Low frequency waves

-

50) **Which of the following is not a guided medium?**

- Twisted pair cable
- Fiber-optic cable
- **Atmosphere** **google**
- Coaxial cable

---

51) As a result of ARPA research, the first network was established which was named as _____.

- INTERNETWORKING
- ARPNET
- **ARPANET** **page 9**
- PACKET NETWORK

---

52) Asynchronous transfer mode (ATM) can dynamically establish a _____.

- **Switched virtual circuit** **page 70**
- Virtual connection identifier
- Virtual path indicator
- None of the given options

---

53) A typical port on an ATM switch operates at _____ or _____ higher.

- OC-2 speed (1000Mbps)
- **OC-3 speed (155Mbps)** **page 72**
- OC-3 speed (100Gbps)

---

54) 10Base-T Ethernet is often called a _____.

- ring shaped star
- ring shaped bus
- **star shaped bus** **page 46**
- star shaped ring

---

55) _____ allows automatic updates.

- **Dynamic routing** **page 62**
- Static routing
- Routing

56) Handling variable length packets leads to memory _____.

- **Fragmentation** **page 72**
- Management
- Location
- Allocation

---

57) In order to forward a copy of frame, the bridge uses _____ address.

- Network
- **Destination** **page 51**
- IP
- Source

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58) We may use _____ to extend local area network.

- Bridge, Router
- Bridge, Hub, Router
- **Bridge, Hub, Repeater** **page 149**
- Bridge, Hub, Repeater, Router

---

59) All of the given options are the types of delay except _____.

- Propagation
- Access
- **Throughput** **page 79**
- Queuing

---

60) While computing shortest path in a graph, next hop information is inserted into _____.

- **Routing table** **page 62**
- Trailer
- Header
- Register

---

61) LAN interface uses _____ to copy frame data directly from main memory

- **DMA** **page 34**
- FDDI

- Hard disk
- Flash

---

62) The Gigabit Ethernet hardware operates at a rate of _____.

- 10 Mbps
- 100 Mbps
- **1000 Mbps reference book page 263**
- 1000 Gbps

---

63) In the early resource sharing era, which resources were shared among users?

- **Computation page 9**
- Floppy Drives
- Hard Disk
- Printers

---

64) If two buildings are located far from each other then bridge, and _____ can be used to connect two LAN.

- **Optical fiber page 51**
- Ethernet modem
- Switch
- NIC

---

65) Which type of network consists of wireless connection of laptops in a university classroom?

- WAN
- MAN
- Fog
- **LAN google**

---

66) A typical bridge has _____.

- One NIC, a CPU a memory and a ROM.
- **Two NICs, a CPU a memory and a ROM. page 50**
- A cache memory, ROM and CPU.

---

67) _____ is used for compressed audio and video, where the data rate depends on the level of compression that can be achieved.

- Constant Bit Rate
- Stable Bit Rate
- **Variable Bit Rate** **page 71**
- Available Bit Rate

---

68) There are _____ types of data stuffing.

- 1
- **2** **page 17**
- 0
- 5

---

69) Dijkstra's algorithm can accommodate weights on _____ in graph.

- **Nodes** **page62**
- Edges
- Path
- Connections

---

70) _____ masks your IP address.

- Firewall
- **VPN**
- Incognito mode

---

71) Static routing computes routing table information at _____ time.

- Load
- **Boot** **page 62**
- Execute

72) _____ was the first automated tool required to see whether the given computer is online or not.

- **None of the above** **page 11**
- Probing program
- PING program
- status program

73) In which type of error detection, the arithmetic sum is calculated before and after the transmission of data at both communication ends?

- CRC
- Party
- **Checksum** **page 19**
- Hash Function

---

74) Computer sends a connection request to the _____ to which it is attached while establishing a switched virtual circuit

- Bridge
- **Switch** **page 70**
- VPI
- Virtual circuit

---

75) The bridges communicate with each other on the network and use _____ algorithm to decide which bridge will not forward frames if a cycle occurs.

- Djikstra
- Shortest path first
- Vector distance
- **Distributed Spanning Tree** **page 53**

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76) Ethernet uses a _____ bit static addressing scheme in which each device is assigned a unique address by the manufacturer.

- 64
- **48**
- 32
- 16

---

77) Weights on graph edges reflect cost of _____.

- Nodes
- **Traversing edge** **page 63**
- Trending edge
- Path

---

78) A _____ relies on the hardware manufacturer to assign a unique physical address to each network interface.

- **Static addressing scheme** **page 34**
- Configurable addressing scheme
- Dynamic addressing scheme
- None of the given

---

79) In _____ topology, if the switch goes down all the computers will be disconnected.

- **Star**
- Ring
- Bus
- Mesh

---

80) Which type of network consists of wired computers in a university classroom?

- MAN
- PAN
- WAN
- **LAN**

---

81) If you as a network administrator want to know the traffic flow of your data while communicating with a remote computer which of the following will be used to know about intermediate routers?

- Ipconfig
- **Ping** **page 11**
- Traceroute
- Arp

---

82) Hardware address is also called _____.

- **Physical address**
- Logical address
- IP address
- Port address

---

83) Thick Ethernet also requires _____ to avoid signal reflectance.

- Transceivers
- AUI cable
- Drop cable
- **Terminators** **page 41**

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84) Which of the following network is using the TCP/IP model globally?

- PAN
- SAN
- LAN
- **The Internet**

---

85) VPNs connections are sometimes called _____.

- **Tunnels** **page 75**
- Open networking
- Routing
- Public network

---

86) To reduce _____ computers can be connected and distributed to a number of hubs, then the hubs can be connected to a switch.

- **Costs** **page 54**
- Quality
- Efficiency
- Noise

---

87) If a _____ is used the communication between two computers will not affect the other pair of computers and they can communicate at the same time

- Hub
- **Switch** **page 54**
- Repeater
- Amplifier

---

88) The length of hardware address is _____.

- **One to six bytes**
- One to five bytes

- One to ten bytes page 34
- One to seven bytes

---

89) A network analyzer is also called _____.

- Network manager
- Network administrator
- **Network monitor page 37**
- Network

---

90) No any error detection scheme is perfect because transmission errors can affect _____.

- Data
- Additional Information
- **Additional information & data google**
- Transmission medium

---

91) The token ring network topology is a ring but wiring topology is a _____.

- **Star page 44**
- bus
- tree
- mesh

---

92) _____ addresses are used in an ethernet frame.

- **IP page 99**
- Port
- Logical
- MAC

---

93) AUI cable connects from NIC to _____.

- BNC Connector
- RJ-45 Connector
- **Transceiver page 41**
- Terminator

94) Public networks are _____.

- More secure than LAN
- **Flexible page 74**
- Limited in boundary
- Intranet

---

95) The maximum cable length of a LAN is fixed because _____.

- Additional hardware are not suitable for the LAN extension.
- **The electrical signal level gets weaker as it travels. Page 48**
- Additional software are not suitable for the LAN extension
- The electrical signal level gets stronger as it travels.

---

96) In _____.network topology is separated from route computation.

- Local area
- **Link-state routing page 64**
- Static routing
- Dynamic routing

---

97) Which layer of the OSI Model directly communicates and controls the transmission medium:

- Transmission
- **Physical**
- Application
- Network

---

98) In _____ technique data is treated as a sequence of integers and their arithmetic sum is computed and the carry bits are added to the final sum.

- CRC
- Bit stuffing
- **Checksum page 19**
- 2D Parity

---

99) A network with _____ jitter takes more time to deliver some packets than others.

- Zero
- **High** **page 80**
- Low
- Moderate

---

100) A network uses _____ to arrange computers in a single closed loop.

- Star Topology
- Mesh Topology
- **Ring Topology** **page 25**
- Bus Topology

---

101) Which of the following tasks is not done by the second layer of the OSI model?

- Flow control
- Framing
- **Channel coding** **google**
- Error control

---

102) ATM is a single technology designed to meet the goals of both _____and _____.

- LAN & PAN
- **LAN & WAN** **page 66**
- LAN & MAN
- MAN & WAN

---

103) LAN and WAN are classified according to their _____.

- **Size** **page 4**
- Connectivity
- Medium
- Mobility

---

104) The identifier that is used for data transfer in the virtual circuit network is called _____.

- **Virtual circuit identifier**
- Global address

- Network identifier
- IP identifier

---

105) In Direct point to point communication adding the Nth computer requires_____ new connections

- N^2
- N-1
- **(N^2 -N)/2** **page 23**
- N (N-1)

---

106) Which of the following is the correct function of bridge?

- MTU path discovery
- Byte stuffing
- Routing information
- **Filtering and forwarding a frame** **page 50, 51**

---

107) Point to point topology is

- Size based
- Mobility based
- **Connectivity based** **page 5**
- Medium based

---

108) IEEE LLC / SNAP header is _____which is used to specify the type of date.

- **8 octets** **google**
- 16 octets
- 32 octets
- 64octets

---

109) IEEE LLC / SNAP header is _____which is used to specify the type of date.

- CRC
- Bit stuffing
- Checksum

110) What is the largest network in existence?

- **The internet** **google**
- A PAN
- A Fog
- Edge Network

---

111) The frame format of any LAN technology must consists of ____ and___-

- Payload and CRC
- Frame header and CRC
- Payload and Frame type
- **Frame header and payload** **page 36**

---

112) _____ can occur a LANs when a packet is delayed because the network is busy

- **Jitter** **page 66**
- Loop
- Deadlock
- Variance

---

113) Which of the following cannot be used to extend a LAN?

- Repeaters
- Routers
- **Fiber modems**

---

114) Links-state algorithm is used in _____

- RIP
- RIPv2
- **OSPF** **page 64**

115) According to size, there are ___classifications of networks

- **2** **page 2**
- 3
- 4
- 5

---

116) In which topology terminators are used at the end of the cable?

- Star
- **Bus**
- Ring
- Mesh

---

117) Switch simulates a _____LAN connecting segments

- Single
- Single shared
- **Bridged** **page 54**
- Complex

---

118) PVS stands for _____

- Permanent virtual circuit page 69
- Private virtual circuit
- Public virtual circuit
- Plastic virtual circuit

---

119) Hardware that calculates a CRC used two simple components _____

- AND unit and XOR Unit
- **Shift register and XOR unit** **page 20**
- Shift register & AND unit
- Shift register and Shift XCR unit

---

120) Thin ethernet is known as _____

- **10 base 2** **google**
- 10 base 5
- 10 base T
- All of the above

---

121) If a sender with a 1 Gigabit NIC and receiver with 100 Megabit NIC wants to communicate with each other than at which speed they can communicate

- **100 Megabit**

- 11 Megabit
- 1 Gigabit
- Gigabit

---

122) _____ is used as an alternative of packet for a specific computer network technology

- Datum
- Token
- Ether
- **Frame** **page 14**

---

123) The sharing of a medium and its link by two or more devices is called ____

- Modulation
- Encoding
- **Multiplexing**
- Line discipline

---

124) The topology each computer is connected to a central hub is called

- Ring topology
- **Star topology** **page 5**
- Tree topology
- Mesh topology

---

125) Asynchronous transfer mode (ATM) is designed to work on fiber but can be used with _____

- Wireless
- **Twisted pair** **page 72**
- Wifi

126) The basic responsibility of NIC is _____

- **To access medium network**
- To resource allocation
- To access memory
- All of the given option

---

127) Frame relay is ____

- **Connection oriented service** **page 64**
- Connectionless service
- Typically ranges for 5 Mbps to 1000Mbps
- Call based

---

128) The length of time required to send a variable length packet is ____

- 5 second
- **Variable** **page 72**
- Constant
- Fix

---

129) ABR stands for _____

- Asynchronous bit rate
- Asynchronous byte recovery
- Asynchronous bit redundancy
- **Available bit rate** **page 71**

---

130) In Token Ring, if all computers are ready to transmit it enforces _____Access

- First come first served
- **Round Robin** **page 30**
- Shorter delay
- Last come first served

---

131) The Ethernet standard specifies that frames are sent using the _____ scheme.

- Differential Manchester
- Not Return to Zero (NR2)
- **Manchester Encoding** **page 27**
- Return to Zero (RZ)

---

132) ATM uses the concept of _____.

- **Connection-oriented** **page 66**
- Connection-less
- variable packet size
- Fixed and variable packet size

---

133) Which of the following is a benefit of point to point communication?

- Quick to configure
- **Security implementation**
- Cost
- Less availability

---

134) _____ is not used to extend LAN.

- Hub
- **Fiber modem**
- Repeater
- Bridge

---

135) Which types of VPNs are used for corporate connectivity across companies residing in different geographical locations?

- Remote access VPNs
- Peer-to-peer VPNs
- Country-to-country VPNs
- **Site-to-site VPNs** **google**

---

136) Star topology is the kind of

- Tree topology
- **Point-to-point topology** **page 05**
- Broadcast topology
- Ring topology

---

137) In a _____ state, a bridge allows simultaneous use of each segment.

- Ready
- **Steady**

- Final
- Startup

---

138) The network occupies larger areas like cities & countries is called

- LAN
- **WAN**
- MAN
- All of the above

---

139) In link state routing _____ send link-state information about local connections.

- Bridge
- **Switches**
- Routers
- Hubs

---

140) Transmission media are usually categorized as _____.

- **Guided or unguided**
- Fixed or unfixed
- Determinate or indeterminate
- Metallic or nonmetallic

---

141) Even parity can be used to check for _____ bit/ bits of errors.

- **1** **page 18**
- 2
- 3
- 0

---

142) Routing table entries can be _____ with a default route.

- **Collapsed**
- Redirected
- Emerged
- Guaranteed

---

143) The system administrator must coordinate to avoid the conflict in_____ hardware addressing scheme.

- Static
- Dynamic
- **Configurable**
- Fixed

---

144) _____ network does not depend on CSMA/CD.

- **Ethernet**
- Fast Ethernet
- Gigabit Ethernet
- Wireless

---

145) Which of the following technologies can extend the diameter of LAN medium?

- Fiber optic
- Repeater
- Diameter of LAN medium is not extendable
- **Both. Fiber optic and Repeaters**

---

146) Connections are formed in Asynchronous transfer mode by starting values in_____ in ATM switches, as opposed to making actual electrical connections.

- Heap
- Stack
- Virtual Circuit
- **Memory locations** **page 67**

---

147) The _____ scheme must be reliable to prevent conflicts.

- Static
- **Dynamic** **page 34**
- Configurable
- Fixed

---

148) Metropolitan area network is used for _____.

- **City** **page 55**
- Building
- Country
- Continent

---

149) Which addressing can be only used in the local area networks?

- **IP** **google**
- Physical
- Port
- Protocol

---

150) The sharing of a _____ among the workers in an office is a goal of resource sharing.

- Mouse
- **Printer** **page 9**
- Keyboard
- Motherboard

---

151) ATM uses fixed size, small cells, 48 octet's data and _____.

- Octets header
- **Octets header** **page 65**
- Octets header
- Octets header

---

152) Which of the followings is not an advantage of framing?

- Missing eot indicates sending computer crashed.
- Missing soh indicates receiving computer missed beginning of message.
- **Delimiters do not contain any user data** **page 16**
- Bad frame is discarded.

---

153) The _____ field of Ethernet frame has the purpose of receiver synchronization.

- **Preamble** **page 36**
- Frame Type
- Data
- CRC

---

154) _____ was the first internet available for the first time.

- ARPANAT
- The ARPNET
- The NET
- **ARPANET** **page 9**

---

155) We need packets rather than bits because _____

- To compress the data
- Packets move faster
- **Sender and receiver need to coordinate to detect transmission errors. Page 14**
- Packets can only move between two terminals

---

156) When an application _____ any data, it makes a copy of the data available to all other computers on the network.

- **Multicast** **page 143**
- Broadcast
- Anycast
- Unicast

---

157) VBR stands for _____

- Virtual Bit Rate
- Virtual Byte Rate
- Variable Byte Redundancy
- **Variable Bit Rate** **page 71**

---

158) Fixed network is a type of networks which is classified with respect to the.....factor

- Size
- Connectivity
- Medium
- **Mobility**

---

159) ------- uses counter rotating rings in which data flows in opposite directions.

- Frame Relay
- ATM
- **FDDI**
- Token Ring
